There are many subtle details in these relations. First, the time domain signal, $x[n]$, is still discrete, and therefore is represented by *brackets*. In comparison, the frequency domain signals, $Re X(\omega)$ & $Im X(\omega)$, are continuous, and are thus written with *parentheses*. Since the frequency domain is continuous, the synthesis equation must be written as an integral, rather than a summation.

As discussed in Chapter 8, frequency is represented in the DFT's frequency domain by one of three variables: k, an index that runs from 0 to $N/2$; f, the fraction of the sampling rate, running from 0 to 0.5; or ω, the fraction of the sampling rate expressed as a natural frequency, running from 0 to π. The spectrum of the DTFT is continuous, so either f or ω can be used. The common choice is ω, because it makes the equations shorter by eliminating the always present factor of 2π. Remember, when ω is used, the frequency spectrum extends from 0 to π, which corresponds to DC to one-half of the sampling rate. To make things even more complicated, many authors use Ω (an upper case omega) to represent this frequency in the DTFT, rather than ω (a lower case omega).

When calculating the inverse DFT, samples 0 and $N/2$ must be divided by two (Eq. 8-3) before the synthesis can be carried out (Eq. 8-2). This is not necessary with the DTFT. As you recall, this action in the DFT is related to the frequency spectrum being defined as a *spectral density*, i.e., amplitude per unit of bandwidth. When the spectrum becomes continuous, the special treatment of the end points disappear. However, there is still a normalization factor that must be included, the $2/N$ in the DFT (Eq. 8-3) becomes $1/\pi$ in the DTFT (Eq. 10-2). Some authors place these terms in front of the *synthesis* equation, while others place them in front of the *analysis* equation. Suppose you start with some time domain signal. After taking the Fourier transform, and then the Inverse Fourier transform, you want to end up with what you started. That is, the $1/\pi$ term (or the $2/N$ term) must be encountered somewhere along the way, either in the forward or in the inverse transform. Some authors even split the term between the two transforms by placing $1/\sqrt{\pi}$ in front of both.

Since the DTFT involves infinite summations and integrals, it cannot be calculated with a digital computer. Its main use is in theoretical problems as an alternative to the DFT. For instance, suppose you want to find the frequency response of a system from its impulse response. If the impulse response is known as an *array of numbers*, such as might be obtained from an experimental measurement or computer simulation, a DFT program is run on a computer. This provides the frequency spectrum as another *array of numbers*, equally spaced between 0 and 0.5 of the sampling rate.

In other cases, the impulse response might be know as an *equation*, such as a sinc function (described in the next chapter) or an exponentially decaying sinusoid. The DTFT is used here to mathematically calculate the frequency domain as another *equation*, specifying the entire continuous curve between 0 and 0.5. While the DFT could also be used for this calculation, it would only provide an equation for *samples* of the frequency response, not the entire curve.